Lab 02 – SD-WAN Foundations

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01. Objective and Topology

Lab 01 steered traffic with policy routes and ended on a finding: a policy route carries no health state. It will keep sending traffic down a path that died, because nothing is measuring the path. SD-WAN is the answer to that, and this lab builds the foundation of it , zones, members, and performance SLAs , on the same topology.

What this lab establishes: a member that fails its health check withdraws its own route automatically, with no link monitor and no administrative action. It also establishes the distinction the rest of the SD-WAN labs depend on , that a link being `alive` and a link *meeting its SLA* are two different states, measured separately and acted on differently.

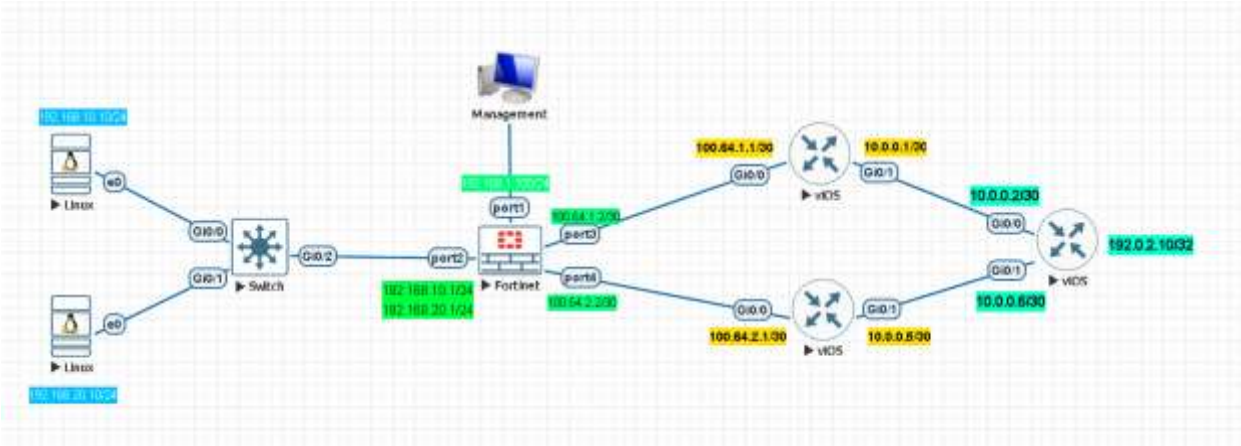


Figure 1 , The topology is unchanged from Lab 01: two upstreams, a shared destination at 192.0.2.10, and two VLANs behind a tagged link.

02. What SD-WAN Is, and What It Needs

Before building anything, it is worth being precise about what SD-WAN actually changes. The name suggests a product; on a FortiGate it is a set of objects that sit between the firewall policy and the routing table, measure every available path continuously, and choose between them per session.

2.1 The problem it solves

A router forwards on destination address alone. Given two internet links, the classic options are all poor:

- **One link, one backup** , a second circuit paid for monthly and idle until the first fails. Failover depends on the interface going down, so a link that stays up while the path beyond it dies is never detected.
- **Both links, equal cost** , traffic is hashed across both. Capacity is used, but nothing measures either path, so half the sessions go into a dead link when one fails. This is the failure documented in section 06 of this lab.
- **Policy routes** , traffic is steered deliberately, which is what Lab 01 built. It works, but the decision is static: the administrator picks the path once, and nothing revisits it when the path degrades.

All three share one deficiency. Nothing in the data path knows whether a link is any good , only whether its own interface is up. SD-WAN closes that gap by measuring each path continuously and making the measurement available to the forwarding decision.

2.2 What it gives you

Benefit	What it actually means on the device
Path awareness	Latency, jitter and packet loss are measured per member on a continuous probe, not inferred from interface state.
Automatic failover	A member that fails its health check withdraws its own route. No link monitor, no scripted action, no administrator.
Quality-based steering	Rules can require a member to meet an SLA before it is eligible, so traffic moves off a degraded link before users notice.
Application-aware routing	A rule can match an internet service or an application, so a specific SaaS can take one path while everything else takes another.
Use of cheap transport	Broadband can carry traffic while it meets the SLA, with the expensive circuit held for when it does not , the normal commercial justification.

Benefit	What it actually means on the device
Simpler configuration	Policies and routes reference one zone instead of one interface each. Adding a third circuit is one member, not a new policy and route set.
Visibility	Per-member quality graphs, SLA pass and fail history, and per-rule member selection are all queryable , the path decision stops being invisible.
Scale	The same object model is what FortiManager templates and SD-WAN overlay orchestration push to hundreds of branches.

The trade-off is worth stating too: SD-WAN adds a measurement layer and a rule table to a forwarding path that previously had neither, so a misconfiguration produces behaviour that no single screen explains. Most of the findings in this lab are examples of exactly that.

▸ 2.3 The five objects

Everything in SD-WAN reduces to five things. If you can name what each one does and in what order they are consulted, most SD-WAN behaviour becomes readable.

Object	What it is
Member	An interface enrolled in SD-WAN , physical port, VLAN, or IPsec tunnel , with a gateway, cost, priority and weight.
Zone	A named group of members. Static routes and firewall policies point at the zone rather than at each interface.
Performance SLA (health check)	A probe sent through each member to a target beyond it, producing latency, jitter and loss figures and a pass or fail against defined thresholds.
Rule (service)	Matches traffic by source, destination, service, internet service or application, and selects members by a strategy.
Implicit rule	Catches everything no rule matched and load-balances it across the zone by the configured mode , source IP by default.

2.4 What has to be configured, in order

The order matters, because FortiOS refuses several of these steps if the previous one is not in place.

#	Step	Why it has to come here
1	Address the WAN interfaces and know each gateway	A member without a gateway cannot be probed or routed through.
2	Remove every direct reference to those interfaces	FortiOS will not accept an interface as a member while a firewall policy or static route points at it directly. This is what section 03 clears.
3	Enable SD-WAN and create the zone	The zone has to exist before members can be placed in it, and before routes and policies can reference it.
4	Add the members, with gateway, cost and priority	These are the paths. Nothing else works until they exist.
5	Create the performance SLA, with at least one SLA target	Without a target, members report no SLA compliance and every SLA-based strategy has nothing to filter on.
6	Add a static route pointing at the zone	Reachability. SD-WAN chooses between paths; it does not create them.
7	Add a firewall policy with the zone as the outgoing interface	A rule can select a member, but the firewall still has to permit the traffic leaving on it.
8	Add SD-WAN rules	Optional. Without any, the implicit rule load-balances everything , which is a working SD-WAN, just not a steering one.

Steps 6 and 7 are the ones people forget

Members and health checks can be configured perfectly and the network still be down, because the zone needs its own route and its own policy , the ones that pointed at the interfaces were deleted in step 2.

Symptom: traffic arrives on the LAN interface and never egresses, while the FortiGate itself still reaches everything. That is section 06 of this lab.

Choosing the probe target

The target must be reachable through every member being measured, and it must sit beyond the part of the path you want to test. A target reachable through only one member will mark every other member dead; a target on the local transit segment proves nothing beyond the first hop.

The opposite risk is concentration: if one probe target becomes unreachable for its own reasons, every member fails at once and the whole zone is withdrawn. Production designs use a highly available target, or two health checks against different targets.

One consequence is worth carrying into every later lab: SD-WAN decides which of the available paths to use , it does not create reachability. A member with no valid route to the destination is skipped silently, however well the rule is written.

▸ 2.8 Where it breaks

Symptom	Usual cause
Interface rejected as a member	A firewall policy or static route still references it directly.
Traffic arrives on the LAN and never egresses	No firewall policy to the zone, or the policy still names the old interfaces.
No default route after building SD-WAN	The static route points at a zone that holds no members, or was never recreated.
Every member reported dead	The probe target is unreachable, or reachable through only one member.
Members healthy but SLA-based steering does nothing	No SLA target defined , sla_map reads 0x0 and every member is treated as failing.
Traffic on the wrong link despite a correct rule	A configured policy route above SD-WAN in the lookup order, or the chosen member has no route to that destination.
Half the hosts fail, the rest work	Equal-cost members with no health check, hashing a share of sessions into a dead path.

03. Clearing the Lab 01 Configuration

SD-WAN cannot take an interface that something else still references. The policy routes, the two default routes and the two firewall policies from Lab 01 all point directly at port3 and port4, so all three have to go before the members can be created.

```
config router policy
    purge
end
```

Figure 2 , Clearing the policy route table. purge empties the whole table on a single confirmation, with no per-entry prompt , take a configuration backup first if the Lab 01 state is still wanted.

The two static default routes and the two firewall policies were removed the same way. From this point until the zone route and policy exist in section 05, the hosts have no path out , which is why the management interface is on port1 and not on one of the VLANs.

04. Zone and Members

A member is an interface enrolled in SD-WAN with a gateway, a cost and a priority. A zone is a group of members that policies and routes point at, so configuration is written once against the zone rather than once per link. Cost is what the Lowest Cost strategy uses in Lab 03; it is an administrative number, not a measurement.

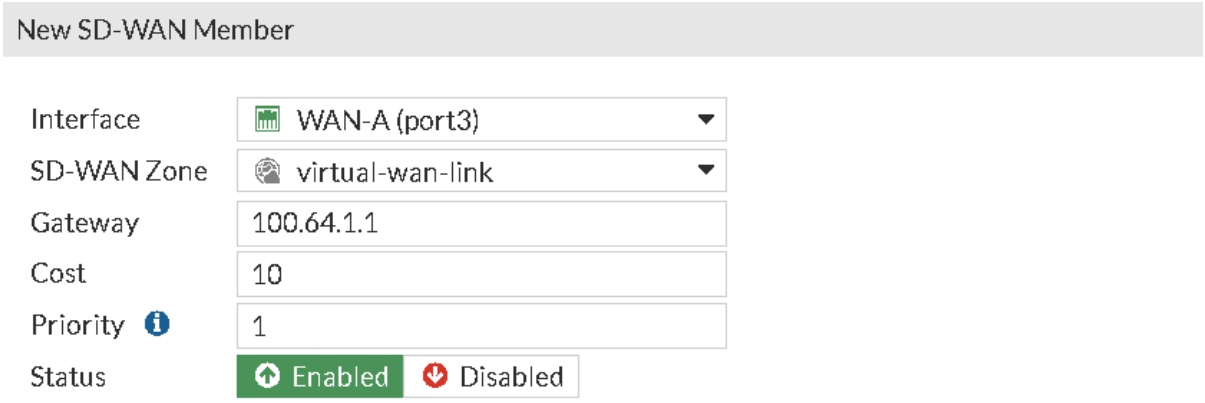


Figure 3 , Adding port3 as a member with gateway 100.64.1.1 and cost 10 , the expensive circuit in this design.



Figure 4 , port4 as the second member, gateway 100.64.2.1 and cost 0 , the cheap circuit.

Member	Interface	Gateway	Cost	Stands in for
1	port3 (WAN-A)	100.64.1.1	10	MPLS , reliable and expensive
2	port4 (WAN-B)	100.64.2.1	0	Broadband , cheap, prefer when good enough

The route and the members must name the same zone

The member dialogue and the static route each name a zone, and they have to agree. A route pointing at a zone that holds no members installs nothing, and the symptom is total loss of outbound traffic with a configuration that looks correct in every individual screen.

Confirm with show system sdwan and get router info routing-table all together , if the zone is right, a default route appears for every member.

05. Reachability Through the Zone

One static route and one firewall policy replace the two of each that Lab 01 needed. This is the architectural argument for SD-WAN in a single before-and-after: the configuration got smaller while gaining the ability to measure and choose between paths.



Figure 5 , A single default route pointing at the SD-WAN zone rather than at an interface and gateway.



Figure 6 , One firewall policy from both VLAN interfaces to the zone, with NAT enabled. Lab 01 required one policy per egress.

	Lab 01 (interface-based)	Lab 02 (zone-based)
Static routes	Two , one per WAN, distances 10 and 20	One , pointing at the zone
Firewall policies	Two , one per egress interface	One , to the zone
Path selection	Policy routes, manually written	SD-WAN members, measured
Health awareness	None without a link monitor	Built in, per member
Adding a third WAN	Another route, another policy, another policy route	One member

06. The Outage During the Build

With the zone route and policy in place the hosts still could not reach 192.0.2.10, while the FortiGate itself could. Working that fault produced the most useful evidence in the lab, so it is documented as it happened rather than tidied away.

```
FortiGate-VM64-KVM # diagnose sniffer packet any 'host 192.0.2.10' 4 20
Using Original Sniffing Mode
interfaces=[any]
filters=[host 192.0.2.10]
18.806837 port3 out 100.64.1.2 -> 192.0.2.10: icmp: echo request
18.808988 port3 in 192.0.2.10 -> 100.64.1.2: icmp: echo reply
19.807156 port3 out 100.64.1.2 -> 192.0.2.10: icmp: echo request
19.809939 port3 in 192.0.2.10 -> 100.64.1.2: icmp: echo reply
20.807457 port3 out 100.64.1.2 -> 192.0.2.10: icmp: echo request
20.808715 port3 in 192.0.2.10 -> 100.64.1.2: icmp: echo reply
21.808031 port3 out 100.64.1.2 -> 192.0.2.10: icmp: echo request
21.809980 port3 in 192.0.2.10 -> 100.64.1.2: icmp: echo reply
22.808505 port3 out 100.64.1.2 -> 192.0.2.10: icmp: echo request
22.810064 port3 in 192.0.2.10 -> 100.64.1.2: icmp: echo reply
```

Figure 7 , A sniffer on the FortiGate during its own ping: echo requests out on port3 and replies back on port3, every time. The path through WAN-A is healthy.

```
97.184267 vlan10 in 192.168.10.10 -> 192.0.2.10: icmp: echo request
98.188572 vlan10 in 192.168.10.10 -> 192.0.2.10: icmp: echo request
99.213025 vlan10 in 192.168.10.10 -> 192.0.2.10: icmp: echo request
100.212657 vlan10 in 192.168.10.10 -> 192.0.2.10: icmp: echo request
101.228636 vlan10 in 192.168.10.10 -> 192.0.2.10: icmp: echo request
```

Figure 8 , The same destination pinged from the Linux host on VLAN 10: no replies at all.

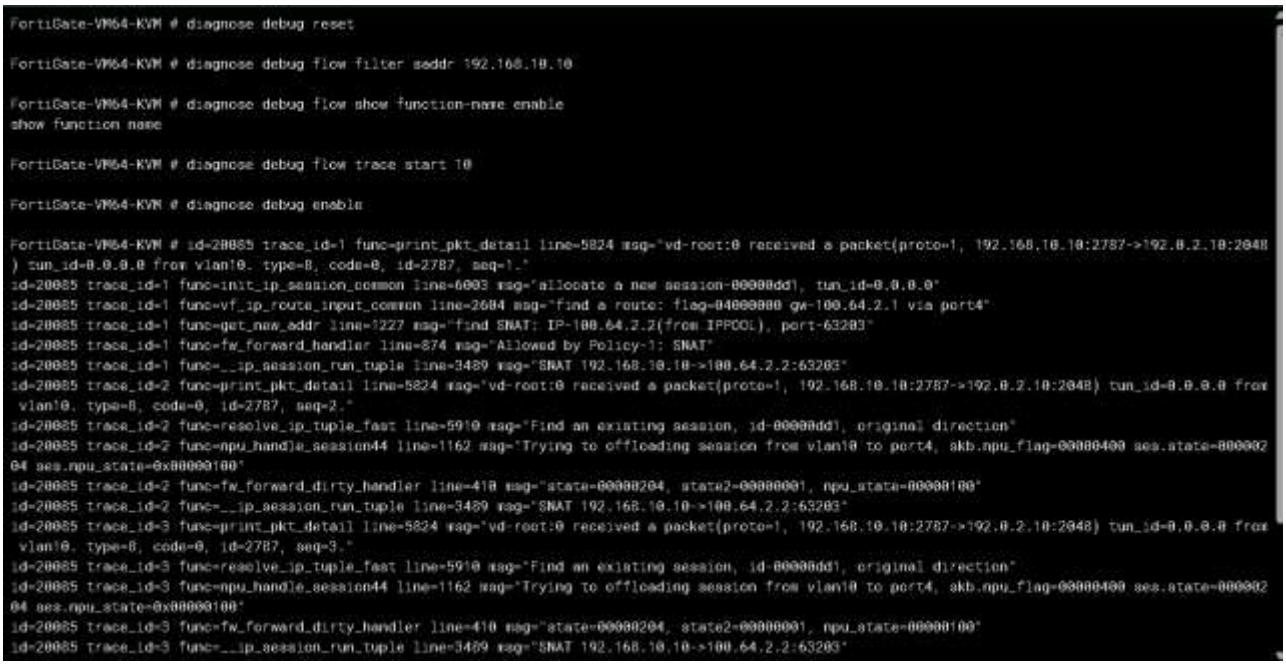


Figure 9 , The flow trace for the host traffic: a route is found via gw-100.64.2.1 on port4, the packet is allowed by Policy-1, and SNAT is applied to 100.64.2.2. Nothing is being dropped , the FortiGate is forwarding it out WAN-B.

That trace is what identified the fault. The FortiGate's own traffic sourced from the port3 address and stayed on WAN-A, while the host's traffic hashed to WAN-B , and WAN-B was dead beyond the FortiGate. One broken link and a load-balancing hash, not two faults.

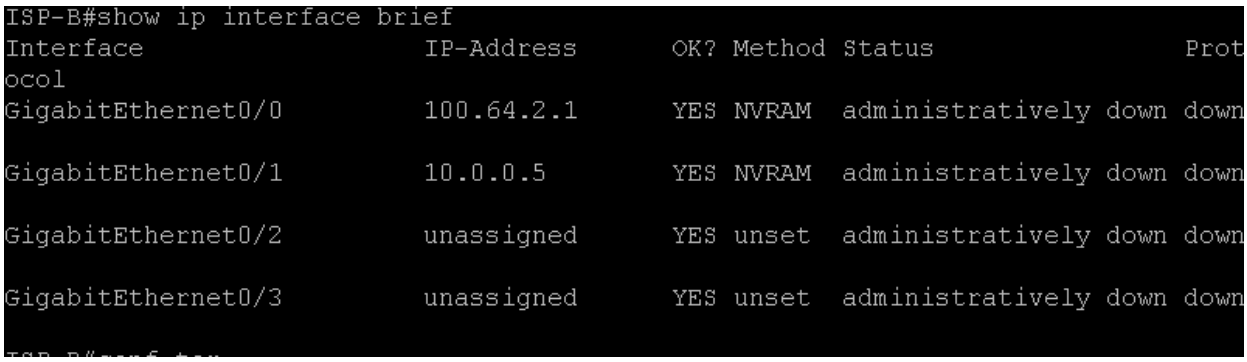


Figure 10 , ISP-B: the interfaces had reverted to shutdown. The running configuration was never written to startup before the lab was stopped.

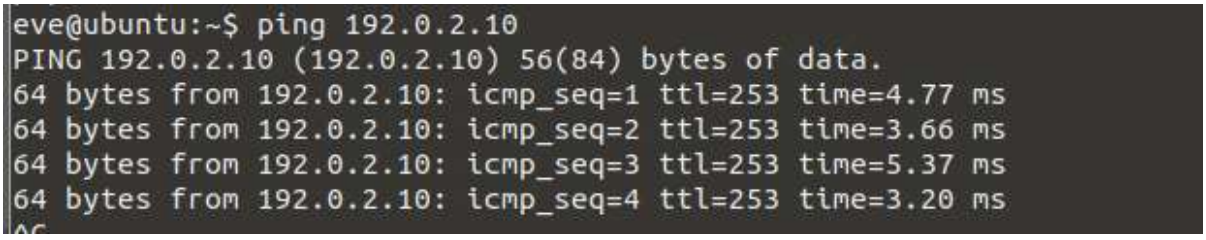


Figure 11 , Connectivity restored from the host once ISP-B was brought back up and its configuration written to memory.

Finding , equal-cost paths with no health check fail silently

Symptom	Hosts on VLAN 10 could not reach 192.0.2.10 while the FortiGate itself could. Routing, policy and NAT were all correct, and the flow trace showed the packet being forwarded normally.
Cause	Both member default routes were installed with equal distance, so the FortiGate load-balanced new sessions across them by source address. Traffic hashed to port4 went into a path that was dead beyond the local gateway. No health check existed yet, so nothing detected or withdrew it.
Fix	ISP-B restored, and a performance SLA created in section 07 so the condition is detected automatically.
Rule	Equal-cost multipath without health checking sends a share of traffic into a dead path and reports nothing. The local gateway being reachable proves only the first hop , a health check must target something beyond it.

Finding , lab routers lose unsaved configuration

Symptom	Interfaces on ISP-B were administratively down after the lab was stopped and restarted, and the earlier interface and routing configuration was gone.
Cause	The running configuration had never been written to startup, so the node booted from an older saved state.
Fix	write memory on every router, and save the lab configurations from the hypervisor before stopping it.
Rule	Treat an unexplained failure after a lab restart as lost configuration until proven otherwise. Check the device that was not part of the last working test , here, every earlier check had exercised the ISP-A side only.

07. Performance SLA

A performance SLA is a probe sent through each member to a target beyond it, producing latency, jitter and packet-loss measurements and a pass or fail against defined thresholds. It is the thing policy routing did not have.

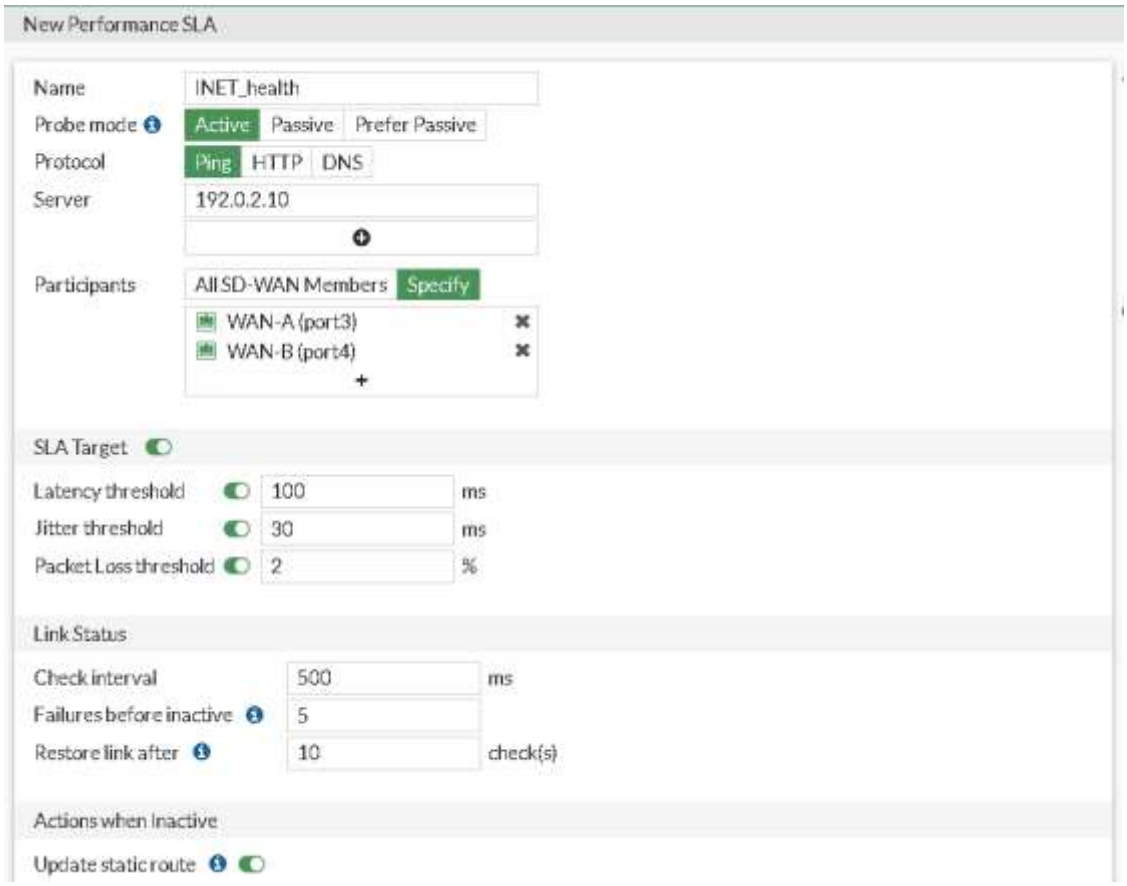


Figure 12 , The INET_health performance SLA: ping to 192.0.2.10 through both members, latency 100 ms, jitter 30 ms and packet loss 2% as targets, five failures to mark a member down and ten successes to restore it.

The probe target matters as much as the thresholds. 192.0.2.10 sits beyond both ISPs, so it is reachable through either member and a failure anywhere along a path is detected. A target inside the lab, or one reachable through only one member, would hide exactly the failure the check exists to find.

```
FortiGate-VM64-KVM # diagnose sys sdwan health-check
Health Check(INET_health):
Seq(1 port3): state(alive), packet-loss(0.000%) latency(2.487), jitter(0.759), bandwidth-up(9999999), bandwidth-dw(10000000), bandwidth-bi(19999999) s
la_map=0x1
Seq(2 port4): state(alive), packet-loss(0.000%) latency(2.809), jitter(0.799), bandwidth-up(10000000), bandwidth-dw(10000000), bandwidth-bi(20000000)
sla_map=0x1
```

Figure 13 , diagnose sys sdwan health-check with both members alive, under 3 ms latency, no loss, and sla_map=0x1 , meeting the SLA target.

```
FortiGate-VM64-KVM # get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default

Routing table for VRF=0
S*      0.0.0.0/0 [1/0] via 100.64.1.1, port3, [1/0]
                  [1/0] via 100.64.2.1, port4, [1/0]
C       100.64.1.0/30 is directly connected, port3
C       100.64.2.0/30 is directly connected, port4
C       192.168.1.0/24 is directly connected, port1
C       192.168.10.0/24 is directly connected, vlan10
C       192.168.20.0/24 is directly connected, vlan20
```

Figure 14 , The routing table with both member defaults installed as equal-cost paths.

Reading sla_map

sla_map is a bitmap of the SLA targets the member currently meets. sla_map=0x1 means target 1 is being met; sla_map=0x0 means no target is met. It is the field to read, because state(alive) only reports whether the probe is answering at all.

The bandwidth figures reported alongside are placeholders , a ping probe measures latency, jitter and loss, not throughput.

08. SLA State Is Not Link State

To separate the two states, the latency threshold was tightened to 1 ms , below what either link can deliver , while leaving both paths perfectly healthy.

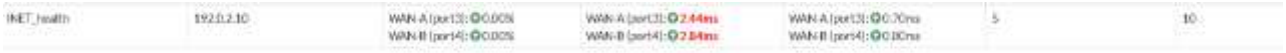


Figure 15 , The GUI performance SLA view: WAN-A at 2.44 ms and WAN-B at 2.84 ms shown in red against the 1 ms target, with 0.00% packet loss on both.



Figure 16 , The same condition on the CLI: both members state(alive) with no loss, and sla_map=0x0 on both.

Both members are up, reachable, and carrying traffic. They simply no longer meet the quality target , alive and SLA-compliant are independent states, and the SLA strategies in Lab 03 act on the second, not the first.

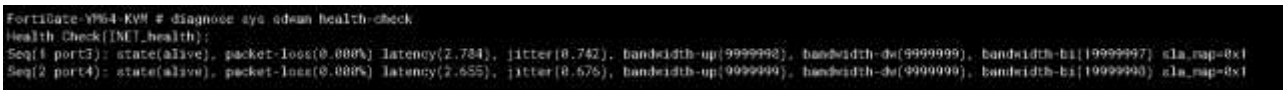


Figure 17 , The threshold returned to 100 ms and both members back to sla_map=0x1.

Finding , a health check with no SLA target looks healthy and steers nothing

Symptom	A health check can report state(alive) with good latency and loss figures while sla_map reads 0x0.
Cause	sla_map reflects the configured SLA targets. With no target defined, or with a target no member can meet, every member reports zero , while every other field in the output looks correct.
Fix	Read sla_map, not state, when verifying. Confirm the config sla block exists under the health check with show system sdwan.
Rule	The SLA-based strategies filter to members that meet an SLA before ranking them. With sla_map=0x0 on every member there is nothing to filter to, so steering silently falls back , a health check that measures correctly but defines no target is worse than none, because every output suggests it is working.

09. Member Failure

The same failure that happened by accident in section 06, now produced deliberately and with a health check watching. Gi0/1 on ISP-B was shut down, so port4 stayed up and its gateway stayed reachable while the path beyond it died.

```
eve@ubuntu:~$ ping 192.0.2.10
PING 192.0.2.10 (192.0.2.10) 56(84) bytes of data.
64 bytes from 192.0.2.10: icmp_seq=1 ttl=253 time=3.99 ms
64 bytes from 192.0.2.10: icmp_seq=2 ttl=253 time=3.67 ms
64 bytes from 192.0.2.10: icmp_seq=3 ttl=253 time=3.07 ms
64 bytes from 192.0.2.10: icmp_seq=4 ttl=253 time=3.25 ms
64 bytes from 192.0.2.10: icmp_seq=5 ttl=253 time=2.66 ms
64 bytes from 192.0.2.10: icmp_seq=6 ttl=253 time=2.46 ms
64 bytes from 192.0.2.10: icmp_seq=7 ttl=253 time=2.65 ms
```

Figure 18 , A continuous ping running from the Linux host before the failure is introduced.

```
ISP-B(config)#interface gigabitEthernet0/1
ISP-B(config-if)#sh
ISP-B(config-if)#shutdown
```

Figure 19 , Shutting Gi0/1 on ISP-B. The FortiGate sees nothing wrong locally: port4 is up and 100.64.2.1 still answers.

```
FortiGate-VM64-KVM # get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default

Routing table for VRF=0
S*      0.0.0.0/0 [1/0] via 100.64.1.1, port3, [1/0]
C       100.64.1.0/30 is directly connected, port3
C       100.64.2.0/30 is directly connected, port4
C       192.168.1.0/24 is directly connected, port1
C       192.168.10.0/24 is directly connected, vlan10
C       192.168.20.0/24 is directly connected, vlan20


FortiGate-VM64-KVM # diagnose sys sdwan health-check
Health Check(INET_health):
Seq(1 port3): state(alive), packet-loss(0.000%) latency(2.317), jitter(1.098), bandwidth-up(9999997), bandwidth-dw(9999998), bandwidth-bi(19999995) sla_map=0x1
Seq(2 port4): state(dead), packet-loss(100.000%) sla_map=0x0
```

Figure 20 , The result: a single default route via 100.64.1.1 on port3, and port4 reported state(dead) with 100% packet loss. The equal-cost pair has collapsed to one path.

No link monitor, no policy route edit, no administrative action. update-static-route is enabled by default on the health check, and the dead member withdrew its own default route. Set against section 06 , the identical physical failure, which went undetected and silently black-holed half the hosts , this is what the lab exists to demonstrate.

Finding , what withdrawal actually reacts to

Observation	Route withdrawal was confirmed under state(dead). The routing table was not captured during the SLA-failure condition in section 08, so whether an alive member that breaches SLA is also withdrawn was not established here.
Why it matters	If withdrawal follows dead state only, an SLA-failing member keeps carrying traffic until a rule with an SLA strategy excludes it , which is the work Lab 03 does. If it follows SLA state too, the routing table alone handles quality.
Next	Capture get router info routing-table all during a tightened-threshold condition, alongside the health check output, and record the answer rather than inferring it.

10. Recovery

Gi0/1 on ISP-B was brought back up and the member allowed to recover on its own.

```
Routing table for VRF=0
S*      0.0.0.0/0 [1/0] via 100.64.1.1, port3, [1/0]
                  [1/0] via 100.64.2.1, port4, [1/0]
C       100.64.1.0/30 is directly connected, port3
C       100.64.2.0/30 is directly connected, port4
C       192.168.1.0/24 is directly connected, port1
C       192.168.10.0/24 is directly connected, vlan10
C       192.168.20.0/24 is directly connected, vlan20
```

Figure 21 , Both member defaults reinstalled as equal-cost paths once port4 recovered.



Figure 22 , The health check immediately after recovery: port4 is state(alive) again, but still carries 39% packet loss and sla_map=0x0, while port3 remains at 0x1.

Finding , a member comes back alive before it comes back compliant

Symptom	After the upstream was restored, port4 returned to state(alive) and its default route was reinstalled, while its packet-loss measurement still read 39% and sla_map stayed at 0x0.
Cause	Packet loss is measured over a rolling window of probes. Immediately after an outage that window still contains the failed probes, so the member answers again long before its measured quality recovers.
Fix	None required , the figure returns to zero as the failed probes age out of the window. recoverytime controls how many consecutive successes are needed before the member is considered up, not how quickly the measurements clear.
Rule	There is a window after recovery in which a member is routable but still failing its SLA. Traffic steered purely by the routing table returns to it immediately; traffic steered by an SLA strategy does not. Designs that need stable failback should expect this gap and size recoverytime around it rather than assuming instant restoration.

11. Command Reference

Command	What it answers
`diagnose sys sdwan health-check`	Per-member latency, jitter, loss, alive/dead state and sla_map
`diagnose sys sdwan member`	Member list with interface, gateway, priority and weight
`diagnose sys sdwan service`	Rule-by-rule member selection (used from Lab 03 onward)
`get router info routing-table all`	Which member defaults are installed, and whether one was withdrawn
`diagnose firewall proute list`	Policy routes including SD-WAN entries, identified by a non-zero vwl_service
`show system sdwan`	Zone, members, health check and the config sla block
`diagnose sniffer packet any 'host x.x.x.x' 4`	Which interface traffic actually leaves and returns on
`diagnose debug flow trace start <n>`	Route lookup, policy match and NAT decision for one flow
`config router policy / purge`	Empties the policy route table in one operation
`execute ping-options source <ip>`	Forces FortiGate-originated traffic onto a chosen source address